Rephrasing dragon wingspan and speed question

Shared by JP

I'm designing an experiment where I'm asking people to interpret data. One question I have is phrased, "When filtering for a single dragon type, how does wingspan relate to flying speed?" I want to change this to be a little more vague, perhaps something like, "For any one dragon type, how does wingspan relate to flying speed?" Can you give me other ways of phrasing that?

Here are several alternative phrasings with varying degrees of vagueness:

Closer to your original (still implies filtering, but softer):

- "Within a given dragon type, how does wingspan relate to flying speed?"
- "Looking at a single dragon type, what's the relationship between wingspan and flying speed?"

More vague (doesn't explicitly cue "filtering" or "single"):

- "For one type of dragon, how does wingspan relate to flying speed?"
- "Considering dragons of the same type, how does wingspan relate to flying speed?"
- "Among dragons that share a type, how does wingspan relate to flying speed?"

Most vague (leaves it open to interpretation whether they should filter, group, or compare):

- "How does wingspan relate to flying speed for dragons of a particular type?"

- "What's the relationship between wingspan and flying speed when you focus on a specific dragon type?"
- "How does wingspan relate to flying speed within dragon types?"

A note on what you're testing: these phrasings differ in how strongly they cue "filter to one category" versus leaving room for someone to look across all types at once, or to mentally group without literally subsetting the data. If part of your experiment is measuring whether people correctly identify that they need to condition on type (rather than looking at the marginal relationship across all dragons), the wording choice matters a lot — "within dragon types" and "among dragons that share a type" nudge toward grouping, while "for a particular type" is more neutral about whether they pick one type or compare across several. If you want, I can also draft a version explicitly testing the opposite framing (pooling across types) as a comparison condition.

The data is a Simpsons paradox, so we're trying to see if people catch the Simpsons paradox or not. So with that in mind, as an experiment design, which do you think makes more sense? Be more vague or less vague?

For a Simpson's paradox detection task, I'd lean toward the more vague phrasing — but with a specific caveat about what "vague" should mean here.

The core thing you're testing is whether people *spontaneously* think to condition on the grouping variable (dragon type) rather than just looking at the pooled relationship. If your question wording already tells them "filter for a single type" or even just "for one type," you've done the conditioning for them — you're not testing whether they catch the paradox, you're testing whether they can compute a within-group correlation once told to look at one group. That's a much easier and different task.

A genuinely vague version would drop the type-conditioning hint entirely and just ask something like:

- "How does wingspan relate to flying speed?"

Then, ideally, you'd pair this with a follow-up or a separate condition that asks the type-specific version ("how does wingspan relate to flying speed within a given dragon type?") to see if responses differ. The gap between the answers to the unprompted question and the prompted one is your real measure of whether people caught the paradox on their own.

A few design options depending on what exactly you want to measure:

- Fully open question, no mention of type at all — see if people notice the subgroup structure unprompted (strongest test of spontaneous detection, but noisier, and some people may not even notice type is a variable in the dataset).
- Open question, but type is visible in the dataset/chart (e.g., color-coded) — see if visual grouping cues are enough to trigger conditioning even without verbal prompting.
- Two-part design: ask the open question first, then explicitly ask the within-type question, and compare answers within the same person.

Your "for any one dragon type" phrasing sits in an awkward middle ground: vague enough that it doesn't feel like an explicit filtering instruction, but still concrete enough that it primes the idea that type matters and that they should pick one type. If your goal is detection, I'd avoid mentioning type at all in the question they have to answer cold, and reserve the type-specific wording for a comparison condition or follow-up.

Our experiment is comparing differences in the user interface of two different types of dashboards. One has manual dials to control what's plotted. The other one just has a simple chat interface where the user can type in the question and the AI decides what chart to make so that they can answer the question. So that is what we're trying assess if or not people think through this more carefully more careful when they are manually clicking versus relying too much on an LLM to decide for them.

That context changes things meaningfully, because now the question wording isn't just a stimulus — it's an input to the AI in one condition, and the AI's interpretation of vagueness becomes part of what you're testing.

A few implications worth thinking through:

In the chat-interface condition, the user types the question, and the AI decides how to chart it. If you give participants a fixed question to type in verbatim, you're really testing how the AI resolves the ambiguity, not how the participant does. A vague question like "How does wingspan relate to flying speed?" might lead the AI to pool across all dragon types into a single scatterplot (likely hiding the Simpson's paradox), while a manual-dial user, just exploring on their own, might stumble into faceting by type or color-coding by type and catch it. That asymmetry could be exactly what you want to capture, but it means the question's vagueness is doing double duty: it affects both what the human thinks to ask, and what the AI decides to chart.

So a few design questions worth pinning down:

Do participants type their own question, or are they given a fixed prompt to enter? If you give everyone the same fixed prompt, you remove a degree of freedom (their own question-asking behavior) but make the comparison cleaner across conditions. If you let them write their own question, you're also capturing whether people think to ask about subgroups in the first place, which adds a layer but also adds noise since now two things vary between conditions (the interface and the question wording).

Are you trying to isolate "people stop thinking when an AI is in the loop" specifically, or are you also interested in "people don't think to ask about subgroups in the first place" regardless of interface? Those are related but distinguishable mechanisms. If you want to isolate the first, you might want the question to be identical and visible in both conditions (e.g., everyone sees "How does wingspan relate to flying speed across dragon types?" displayed on screen, and dial-users have to manually build a chart to answer it, while chat-users type it in and the AI charts it for them). That way the same ambiguity is present in both conditions, and the dependent variable is purely about exploration behavior and final interpretation, not about who generated the question.

If instead you want to test whether the AI's framing of the answer is what lulls people, you might intentionally have the AI's chart pool across types (a realistic failure mode) and see whether people accept that chart's apparent conclusion without pushing back, versus whether dial-users discover the subgroup structure through their own manual exploration.

Given that framing, I'd lean toward giving everyone the same fixed, genuinely vague question (something like "How does wingspan relate to flying speed?") rather than varying the wording by condition. That way, vagueness is held constant, and what differs is only how each interface handles that vagueness, and whether the human catches a misleading or pooled answer. If you want, I can help sketch out the specific charting behavior you'd expect from a "naive" AI assistant in this scenario, so you can decide whether to script that behavior deterministically rather than relying on an actual LLM in the loop (which would add a lot of variance to your experiment).

We're still trying to figure out exactly what questions we want to answer. So let's think about this in a planning mode and think through the pros and cons of different experiment designs. We really have two key levers about the user interface for the dashboard: one is manual with buttons, and the other is a chat interface. We've already set it up where users have a simple data set and five basic questions about the data and users will be split into two groups, one manual, one with a chat interface. That is already implemented. And the

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There's a lot of good material here, and I think the most useful thing I can do first is point out that you actually have two distinct research questions tangled together, because they pull the design in different directions.

The two questions hiding in your setup

The first is a *cognitive offloading* question: does handing the work to an AI make people disengage, so they retain less and reason less deeply even when the immediate answer is fine? Your recall-at-one-hour measure is the signature test for this, and it's backed by real literature — the generation effect and "desirable difficulties" both predict that people who build the chart themselves (manual) will remember the data better an hour later than people who had it handed to them (chat), *even if immediate accuracy is identical*. That's a clean, theory-grounded prediction, and recall is where I'd expect your UI effect to be largest.

The second is an *inferential-trap* question: does the interface affect whether people catch a misleading pattern? That's what the Simpson's paradox tests. It's related ("are people thinking carefully?") but it's measured differently and, crucially, it's confounded in a way recall is not (more on that below).

These can coexist in one study, but trying to fully power both is where studies turn to mush. It's worth deciding which is primary.

Lever by lever

Simpson's paradox. Its value is that your five "basic" questions probably suffer from ceiling effects — both interfaces can answer "what's the max flying speed?" and accuracy variance collapses. The paradox is a discriminating item that puts variance back in. The catch is that it's a single, near-binary item, so as a powered continuous DV it's weak; I'd treat it as a focal confirmatory item rather than something you compute a tidy effect size on. And it only works if the manual UI makes grouping-by-type *discoverable but not automatic* — if there's no "group by type" control, manual users can't catch it either and the item tests nothing; if the chart auto-facets, everyone catches it and again nothing. The paradox has to sit in that middle zone where catching it reflects user initiative.

The AI-behavior confound (this is the sharp one). In the chat condition, whether the paradox gets caught is mostly a property of the AI, not the participant. If your AI facets by type, chat users catch the paradox *for free* and may beat manual users. If it pools, they get misled. So the direction of your headline effect on the paradox item is set by how you configure the AI. This means you should not "just use an LLM" live — that injects huge variance and makes the result about your model's mood that week. Script the AI's chart outputs deterministically (Wizard-of-Oz / pre-baked responses), and decide deliberately that for the paradox question it produces the *pooled, misleading* chart. Then the paradox becomes a genuinely interesting automation-complacency test: do manual users discover the subgroup structure through their own exploration more often than chat users think to question the AI's pooled chart? That's a clean story. But it depends entirely on controlling the stimulus.

Explainer button. Its most interesting use isn't as a generic helpfulness feature — it's as a *mechanism test* for the offloading hypothesis. Putting the explainer on the manual condition asks: does adding AI assistance to manual users erode their advantage? If manual-without-explainer beats both manual-with-explainer and chat (which sit at the same lower level), you've shown it's the AI involvement per se, not the clicking, that drives offloading. That's a strong mechanistic result. The cost is that it makes you a 2×2, and the explainer's meaning differs across UI conditions (in chat it explains the AI's own chart, which is partly redundant; in manual it explains the user's chart), so the chat-side cell is muddier.

Power, and why factors are expensive

Rough anchors for between-subjects, 80% power: a medium main effect (d≈0.5) needs ~64/group, ~128 total; a smaller, more realistic effect (d≈0.3) needs ~175/group, ~350 total. The trap is interactions. A 2×2 interaction of the same raw magnitude as a main effect typically needs about four times the total N to detect — so a UI×Explainer interaction can push you toward 500–700+. Then your one-hour recall introduces attrition; if 30% don't return, inflate by ~1.4×. And three DVs (accuracy, speed, recall) means either a designated primary DV or a multiple-comparison penalty.

The honest workflow is: pick one primary DV and one primary contrast, state the smallest effect that would matter to you, power for *that*, and label everything else exploratory. Trying to power the interaction *and* the paradox *and* three DVs simultaneously is what blows up your N.

One free lever worth noting: your five questions are already within-subjects, which is efficient. UI realistically can't be within-subjects (once someone's seen the data, you can't re-test them on it with the other interface unless you build a second parallel dataset and counterbalance — doable, and it would slash your N, but it risks a dataset-difficulty confound).

Two concrete candidates

The cheap, high-value design keeps your existing two groups, adds nothing structural, but makes one or two of the five questions paradox traps with a deterministically-scripted (pooled) AI chart. Primary DV: one-hour recall, primary contrast: manual vs chat. Paradox accuracy is a focal secondary item. This is well-powered at ~150–350 total depending on your effect size, tests the offloading question cleanly, and uses the paradox as a confirmatory probe. The cost is you don't learn anything about the explainer.

The ambitious design is the 2×2 (UI × explainer) to get the mechanism test, with the paradox embedded as before. This answers the more interesting "is it the AI or the clicking?" question but roughly triples your recruitment and asks you to power an interaction. I'd only commit to this if the offloading mechanism is the actual point of the paper.

The single decision that drives all of this: is your headline claim about *retention/engagement* (recall, offloading — then keep it simple and make recall primary) or about *being misled by a tool* (the paradox and automation complacency — then the controlled-AI paradox item becomes central and the explainer 2×2 is tempting)? Which of those two is the result you'd most want to be able to state in the abstract?

One wrinkle is the ambiguity of the Simpsons paradox question. If we just phrase it in a vague way, like what's the relationship between these two variables, it's unclear whether or not the user will have detected that there is a different answer when you're answering for a single dragon type versus all of them combined. We could pre-prompt it in the AI chart to always show the combined figure And that way you could see in their chat history if the user followed up to ask about a specific single type. But it still doesn't help the fact that you could answer that question either way and still be correct. So that makes it hard to assess was this correct. One workaround is maybe to have a follow-up question that just asks how did you come to this answer. and we could see if they reasoned through that way, but I'd like your thoughts on this issue.

The unscorability is actually a useful signal: it's telling you that the descriptive question doesn't match the construct you care about. "What's the relationship between these two variables?" has two defensible answers because the relationship genuinely is positive within type and negative pooled — so a person can be fully paradox-blind and still land on a "correct" answer by luck of which view they happened to look at. The thing you actually want to measure isn't "positive or negative," it's "did they recognize that the answer depends on the level of aggregation." Once you frame it that way, the fix is to stop asking a descriptive question and ask one where only paradox-aware reasoning produces the right answer.

The decision-question reframe. Simpson's paradox is famous precisely because there's usually a decision or causal question where the pooled answer leads you astray (Berkeley admissions, the kidney-stone treatment example). Borrow that structure. Instead of "what's the relationship," ask something like: "If you wanted to breed a dragon that flies faster, should you select for larger or smaller wingspan?" The person misled by the pooled chart says "smaller" (pooled negative); the person who caught the reversal says "larger" (positive within type, which is the action-relevant relationship). Now you have an objectively scorable answer *and* you haven't cued them that type matters — the word "type" never appears. People who got fooled by the pooled chart will confidently give the wrong answer, which is exactly the behavior you're hunting for.

One important caveat with this approach: it only has an unambiguous correct answer if your cover story makes dragon type a *confounder* — something that independently affects both wingspan and base flying speed — rather than a *mediator* on the causal path from wingspan to speed. If type sits on the causal path, the pooled answer can actually be the correct one to act on, and you've reintroduced the ambiguity at the causal level. Since it's a synthetic dataset you control the narrative, so just make sure the framing establishes type as a common cause. This is worth getting right because a surprising number of published Simpson's examples are sloppy about exactly this.

On your follow-up "how did you come to this answer" idea. It's worth including, but as a secondary/process measure, not as the thing that rescues scoring. Its weaknesses: it invites post-hoc rationalization (people confabulate clean-sounding reasoning they didn't actually use), it carries demand characteristics (asking how they reasoned signals that the reasoning was supposed to be careful, which can prompt people to retroactively "notice" the subgroups), and free-text reasoning needs a coding rubric with two raters and an inter-rater reliability check, which is real labor. Where it earns its keep is triangulation: if someone gives the right decision answer *and* spontaneously mentions the reversal in their reasoning, that's much stronger evidence of true detection than either signal alone.

Make the behavioral trace symmetric. Your "pre-prompt the AI to show the combined chart, then watch for a follow-up" instinct is good, but it only works as a clean comparison if the manual condition has a parallel default and a parallel affordance. So: have *both* interfaces default to the pooled view (AI shows pooled; manual's initial chart is pooled), and make sure both have a way to disaggregate (AI via a follow-up question; manual via a "group/split by type" control). Then detection-by-exploration becomes the same operationalization in both arms — *did the user deviate from the pooled default to look at subgroups* — and you can compare that rate across UI conditions without an apples-to-oranges problem. If manual defaulted to faceted while chat defaulted to pooled, you'd be measuring your own design choices, not the participants.

Add confidence. Capture a confidence rating on the decision question. The signature of automation complacency isn't just being wrong — it's being wrong *and* confident. If your chat users who never disaggregated give the wrong answer with high confidence while manual users who explored give the right answer, that confidence-accuracy dissociation is a more compelling result than the accuracy difference alone, and it's nearly free to collect.

A note on the spontaneous-vs-prompted tension. There's an unavoidable tradeoff lurking here. Spontaneous detection (just show them a vague question and see if they catch it) has high construct validity but is noisy and hard to score. Prompted detection (explicitly ask the within-type and pooled versions side by side) is cleanly scorable but cues the structure, so you're no longer measuring whether they'd have caught it on their own. The decision-question approach is attractive specifically because it threads this needle — scorable without cueing. If you ever do want the explicit two-part comparison, the way to keep it fair is to present it identically in both UI conditions; the cueing inflates everyone's absolute detection rate, but since it's held constant across arms, the *between-condition* comparison you care about stays valid. You'd just be honest in the writeup that you're measuring prompted rather than spontaneous detection.

If it's helpful, the cleanest version of all this is probably a short staged sequence on the paradox item: the decision question first (uncued, scorable, with confidence), then the free-text reasoning, then optionally a direct "does this relationship change depending on whether you look at one type or all of them together?" as the most-cued backstop. That ordering lets you see how much scaffolding each person needed before they caught it, which turns a fragile binary into something closer to an ordinal measure — and gives you several converging signals instead of betting everything on one unscorable question. Want to work through what the causal cover story for the dragon data should be so the decision question has a defensible right answer?

I like the idea of phrasing the question in a way that forces you to rationally think about the result from breeding a dragon. That will help with the queuing problem. For context, how we're setting up the experiment is to start with a scatterplot that is pooled for both groups. So the AI version will always show a pooled scatterplot. It'll be pre-prompted to do so when this question comes up. And the manual version will have each species of dragon as checkboxes alongside another checkbox called "all." And the "all" button will be selected

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The phrasing fix you liked solves cueing *inside* the paradox item, but the thing I'd flag hardest is that the other four questions can leak the paradox *around* it. Look at the set as a block from the participant's point of view:

- Q2 explicitly frames a relationship "across all combined."
- Q4 explicitly frames the same kind of relationship "filtering to a single type."

Those two questions, sitting in the same instrument, essentially teach the concept you're trying to catch people *failing* to grasp. A participant who'd never have thought to disaggregate on their own gets handed the idea that "combined" and "single type" are different lenses that can yield different answers. So by the time they reach your carefully uncued breeding question, they may answer it correctly not because the interface made them reason carefully, but because Q2/Q4 primed the contrast. Whatever order they're in, the breeding question's uncued property is fragile to its neighbors.

The habituation confound, which is sharper and asymmetric. Q1, Q3, and Q5 can't be answered from the pooled default at all — you *have* to look per-type. In the manual condition, that means the user clicks off "all" and selects individual types, building the exact disaggregation habit that catching the paradox requires. In the chat condition, the user asks a type-specific question and the AI just hands back a per-type chart — they receive the breakdown without ever forming the habit of suspecting the pooled view. So the warmup questions train your manual group to deviate from pooled and *don't* train your chat group the same way. If the paradox item comes after them, a manual advantage on detection could be an artifact of question order rather than a genuine property of the interface. That's a confound that would directly attack your headline claim.

Two consequences follow, and they're the main design levers:

There's really only one clean spontaneous-detection slot, and it has to come first. The "did they deviate from the pooled default" trace is only a pure measure of unprompted detection on the *first* question that rewards deviating. After that, deviation is habituated for the manual group and pre-served for the chat group. So the breeding/paradox question should be Q1 of the sequence — before anything that requires per-type views. Everything after it can be lookups and prompted probes.

You probably want one deliberately non-reversing relational item. If every relationship question is a trap, a sharp participant learns the trick ("always check subgroups") and your paradox item stops measuring detection and starts measuring whether they cracked the experiment's pattern. A relationship that's the *same* sign pooled and within-type is a valuable foil — it shows you who's reasoning versus who's mechanically applying "disaggregate everything." Q2 (weight × speed) is the natural candidate for this role, *if* you design that relationship to be stable. Right now I can't tell whether weight × speed is also paradoxical in your synthetic data; you should decide that on purpose, because if it reverses too, "across all combined" has the same unscorable ambiguity Q4 had.

Going question by question with that framing:

Q1 / Q3 (highest avg speed, largest avg wingspan). These are fine as lookup warmups and they're good recall items (memorable categorical answers). Their real job in this design is the offloading metrics — you'd expect ceiling on accuracy but plausibly a speed difference and, most interestingly, a recall difference an hour later. Just know they're disaggregation-trainers, so they belong *after* the paradox item, not before. Phrasing is already clean; I wouldn't touch them except to confirm the "All same" distractor is genuinely wrong in your data.

Q2 (weight × speed). Decide its role first. As a non-paradoxical foil, drop "combined" — that word both pre-pools the answer for them (removing any reasoning) and reinforces the combined/single contrast you're trying not to telegraph. "How are weight and flying speed related?" with the same four options is cleaner, and if the relationship is stable across aggregation levels it's honestly scorable regardless of how they looked at it. If instead weight × speed *does* reverse, then it's a second paradox and you've got the same ambiguity problem — in which case either fix the data so it's stable, or convert it into a second decision-framed question.

Q4 (wingspan × speed within type). This is the cued descriptive paradox, and "filtering to a single dragon type" is exactly the giveaway you already decided to remove. Replace it with the breeding decision version, something like: *"A breeder wants dragons that fly as fast as possible. Should they breed for larger or smaller wingspans?"* — options Larger / Smaller / Wingspan doesn't affect speed / Cannot determine. The pooled-misled answer is "smaller," the paradox-aware answer is "larger," and the word "type" never appears. This only has a defensible right answer if your data narrative makes type a *confounder* (a common cause of both wingspan and base speed) rather than something on the causal path from wingspan to speed — worth locking down before you finalize.

If you still want the explicit descriptive contrast (the old Q2/Q4 pair), keep it but move it to the *end* as a prompted backstop, and read it as "prompted detection" rather than spontaneous. The gap between the uncued breeding answer (early) and the cued within-type answer (late) is itself an interesting per-person measure of how much scaffolding they needed.

Q5 (wingspan at same claw length, Mountain vs Sea). This is a different and rather nice construct — conditioning on a third variable, the "hold X constant and compare" move — so it complements the paradox rather than duplicating it. Two cautions. First, it's only scorable if the synthetic data actually makes it determinable, which requires the claw-length ranges of Mountain and Sea dragons to *overlap* (you can't compare "at the same claw length" if the two types occupy disjoint ranges). Second, it names specific types, so it's another disaggregation-rewarding item and belongs after the paradox slot. Phrasing is good as written; "Cannot determine from the data" is the right honest-uncertainty distractor.

Pulling it together, a sequence that protects your focal measure:

1. **Breeding decision question** (uncued paradox, scorable) — first, with a confidence rating attached, then the free-text "how did you decide." This is your one clean spontaneous-detection slot and your behavioral-trace slot (did they deviate from the pooled default before answering).
2. Q5-style conditioning question — second probe of "do they think to control for something."
3. Q1, Q3 lookups — offloading/recall workhorses.
4. Q2 as a stable foil.
5. Optionally the explicit within-type/pooled descriptive pair as a prompted backstop, read as prompted not spontaneous.

Two carry-overs from before that are nearly free and strengthen the story: attach a confidence rating to the focal question (the compelling result is *wrong-and-confident* among chat users who never disaggregated, versus the manual users who clicked off "all"), and keep the wording byte-for-byte identical at the one-hour recall phase or the recall comparison breaks.

The one thing I'd want you to pin down before phrasing anything else, because it determines whether two of these questions are even scorable: in your synthetic data, which relationships *reverse* under aggregation and which stay stable? You have four variables in play — speed, wingspan, weight, claw length — and right now it's implicit which pairs are traps. Do you already know the intended sign (pooled vs within-type) for weight × speed and for the claw-length comparison, or is that still open?

We created the dataset ourselves. It's a synthetic dataset. So the only one that has the paradox is the wingspan versus speed. The other ones are intuitive. Like, larger dragons are heavier and have bigger claws.I agree that the paradox question should come first, so we'll put that one first, and then we can put any of these other questions. The goal is just to have a few extra questions so we can assess accuracy. No real tricks are needed for the other questions.

Your ordering instinct is right, and putting the paradox first actually resolves the habituation confound I was worried about — once it's first, nothing before it can train the manual group to deselect "all." So that's clean. But making the other questions trick-free surfaces two new things worth being deliberate about.

The paradox item is now also the first-contact item. Whichever interface someone's in, the breeding question is the very first thing they do, so detection is entangled with UI novelty — the manual user is simultaneously discovering what the checkboxes do and deciding whether to use them. That noise works against you precisely on your focal measure. The fix is a practice question before the scored sequence, but it has to be answerable from the pooled default (something like "how many dragons are in the dataset?" or a single pooled lookup) so it teaches the mechanics without teaching disaggregation. If the practice item requires per-type views, you've recreated the exact training confound you just avoided by reordering.

"Intuitive" splits into two very different kinds of filler, and only one is useful. There's a real distinction hiding in your set:

- *Arbitrary categorical lookups* — "which type has the highest average flying speed," "largest average wingspan." The answer (Forest vs Mountain vs Sea) can't be guessed from priors; you have to read it off the data. These are your good filler: genuine recall items at one hour, and they still carry some accuracy signal.
- *Directional relationships that match common sense* — "bigger dragons are heavier," "bigger claws." These are answerable without ever touching the dashboard. That's a double problem: accuracy sits at ceiling so it won't distinguish the interfaces, *and* one-hour recall is contaminated because people "remember" the answer from intuition rather than from having seen the data. So they don't measure what you want them to.

The consequence for your DVs: with only one paradoxical item and the rest trick-free, accuracy's discriminating power is almost entirely concentrated in the paradox question. Your interface-difference signal really lives in three places — the paradox item, time-to-answer, and recall on the arbitrary-mapping items. Worth planning your power around those specifically, and not being alarmed when accuracy on the intuitive items shows null effects across interfaces. That's the expected result, not a failure of the study; it just means those items are doing recall/speed work, not accuracy work.

Q5 doesn't actually belong in the "trick-free" bucket. "At the same claw length, which type has a larger wingspan" is a control-for-a-third-variable reasoning task — conceptually it's a cousin of the paradox, not an intuitive lookup. So either keep it on purpose as a second, different-flavored reasoning probe (it's a nice complement, since it tests "hold something constant" rather than "disaggregate"), or swap it for a clean categorical lookup if you really want the non-focal items trick-free. And it's only scorable if the claw-length ranges of Mountain and Sea dragons overlap — if they're disjoint, "at the same claw length" has no answer.

One small data check: confirm weight × speed is genuinely unambiguous in your synthetic data. "Heavier" doesn't obviously imply faster or slower (more mass vs. more power), so that directional question may be less intuitive than the size→weight and size→claw ones, and you don't want a filler item that's accidentally arguable.

So a minimal, clean version: a pooled-answerable practice item, then the breeding question (with confidence rating and the free-text "how did you decide"), then your arbitrary-mapping lookups for recall, with Q5 retained only if you're treating it as a deliberate second reasoning probe. Do you want to keep Q5 as that second probe, or would you rather the only reasoning item be the paradox so the contrast is as clean as possible?

Keep question five as the second probe and then give me the five questions in the order that you would recommend include the question and the selection responses.

Here's the full set in recommended order, with response options:

Q1 — Paradox item (uncued, scored as your focal measure). Attach a confidence rating and the free-text "how did you decide" follow-up to this one only.

> A dragon breeder wants to produce dragons that fly as fast as possible. Based on the data, should they breed for larger or smaller wingspans?
> - Larger wingspans *(paradox-aware / correct)*
> - Smaller wingspans *(pooled-misled)*
> - Wingspan doesn't affect flying speed
> - Cannot determine from the data

Q2 — Second reasoning probe (control-for-a-third-variable).

> At the same claw length, which dragon type tends to have a larger wingspan — Mountain Dragons or Sea Dragons?
> - Mountain Dragon
> - Sea Dragon
> - Both have the same wingspan
> - Cannot determine from the data

Q3 — Arbitrary categorical lookup (recall + accuracy workhorse).

> Which dragon type has the highest average flying speed?
> - Forest
> - Mountain
> - Sea
> - All the same

Q4 — Arbitrary categorical lookup (recall + accuracy workhorse).

> Which dragon type has the largest average wingspan?
> - Forest
> - Mountain
> - Sea
> - All the same

Q5 — Directional relationship (filler / accuracy).

> How are weight and flying speed related?
> - Positively related
> - Negatively related
> - No relation
> - Cannot determine from the data

A few notes carried over from the design discussion, so the ordering does what it's meant to:

The paradox question has to be literally first, before any practice item that rewards disaggregation — otherwise you reintroduce the habituation confound. Precede the whole set only with a pooled-answerable practice item (e.g., a simple count or single pooled lookup) so people learn the controls without learning to deselect "all."

Two scoring dependencies to confirm in your synthetic data before locking these: Q2 is only answerable if Mountain and Sea claw-length ranges *overlap* (otherwise "at the same claw length" has no defined answer), and Q5's weight×speed sign should be unambiguous — "heavier → faster or slower" isn't as intuitively obvious as size→weight, so make sure the data commits clearly to one direction or that item turns arguable.

I dropped weight×speed to last because it's your weakest discriminator (closest to guessable from priors), so it's pure filler; Q3 and Q4 are the better non-focal items since their answers can't be intuited and they make genuine one-hour recall probes. And keep all wording byte-for-byte identical at the recall phase or the recall comparison breaks.